

A vibrant explosion of pink, orange, and blue powder against a black background. The powder is captured in mid-air, creating a starburst effect with various shades of magenta, fuchsia, and cyan. The explosion is centered, with the colors radiating outwards in all directions.

ECON 2041

Introductory Econometrics

Intro to unit

Outline

- Intro to teaching team
- Salary prediction game
- Technology and learning materials
- Assessment
- Learning outcomes

Teaching team

Unit convenor: Associate Professor Emilia [Tjernstrom](#)

Email: emilia.tjernstrom@mq.edu.au

Consultation hours: Tuesdays 9 - 10 (both in-person and Zoom)

Tutors:

- Fazeel Jaleel
- Syed Atif
- Adithya Krishna

Yes, I know...

It's a beast!

~*churn*-strom

Tutorials start in week 2

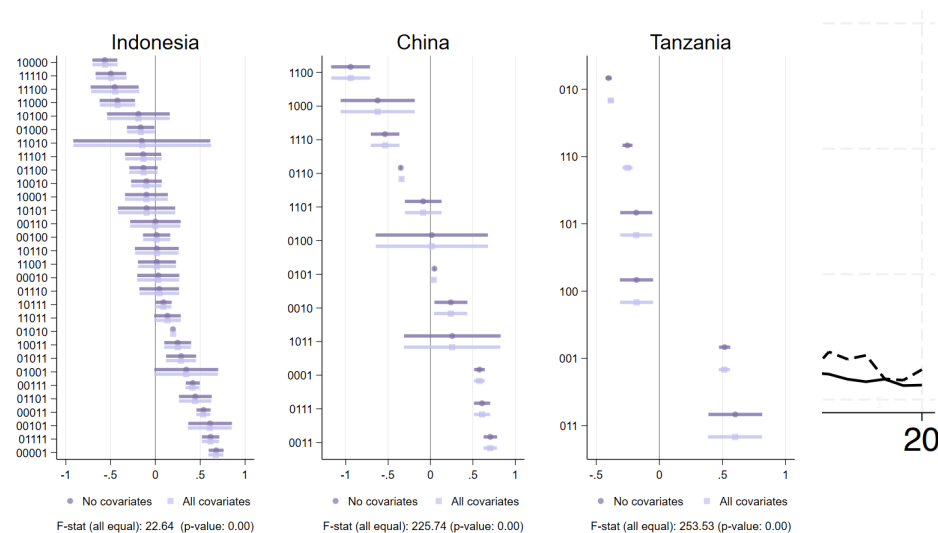
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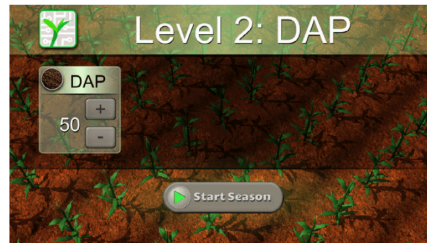
A little more about me:



I study how policies work & for whom

I apply and develop econometric tools to understand treatment effect heterogeneity

Teaching team



(a) Input selection



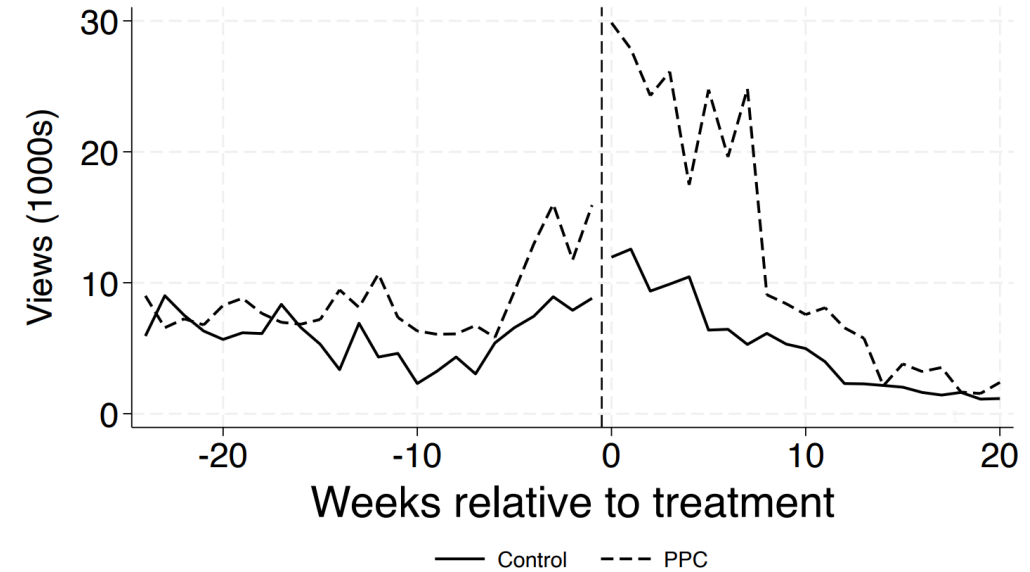
(b) Fertilizer application



(c) Seed planting



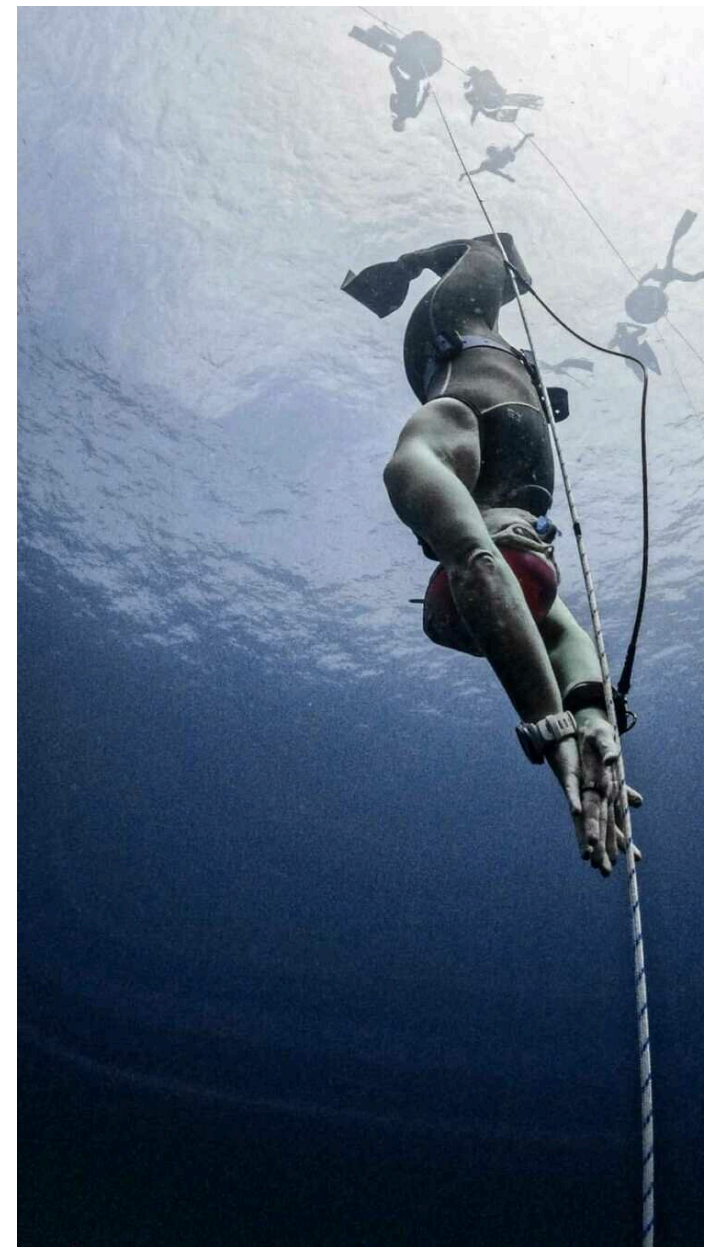
(d) Harvest summary screen



I design & run large-scale RCTs in developing countries



A little more about me:



The Graduate Outcomes Survey (GOS)

Where I got the data:

- A national survey of new graduates, run every year since 2016
- In 2025: over 120,000 graduates from 136 institutions, including all 42 Australian unis
- Graduates respond 4–6 months after finishing, so these are first-job salaries
- We will look at the median annual salary of grads working full-time

Remember, the median tells you what the *typical* new graduate earns, not the average

Salary prediction game

Ten fields of study, for each one: what do you think the typical new graduate earns per year?

1. Open your portal link (in welcome email)
2. OR scan the QR code & enter your ID
3. Move the sliders to your best guess



Quantitative skills pay

Evidence from Texas: used AI to classify coursework for every public-university student, linked to their quarterly earnings after graduation

- +1 quant course: ca. 1.1% higher early-career earnings, roughly \$150 more per quarter
- Writing-focused coursework: no detectable earnings return
- Largest gains where the skills are scarcest:
 - 10.6% vs 1.9% for students in less-quantitative majors
 - 9.8% vs 4% for URM students (per SD of quant coursework)

This unit is a great place to start building these skills...

By the end, you will have a portfolio of notebooks showing off the skills you've learned!

Logistics & plan

Software

Excel

You can get Excel & other MS Office products through your MQ account

Zoom

- How to use it (remember to test your mic)
- We expect people to have their cameras on when in Zoom meetings

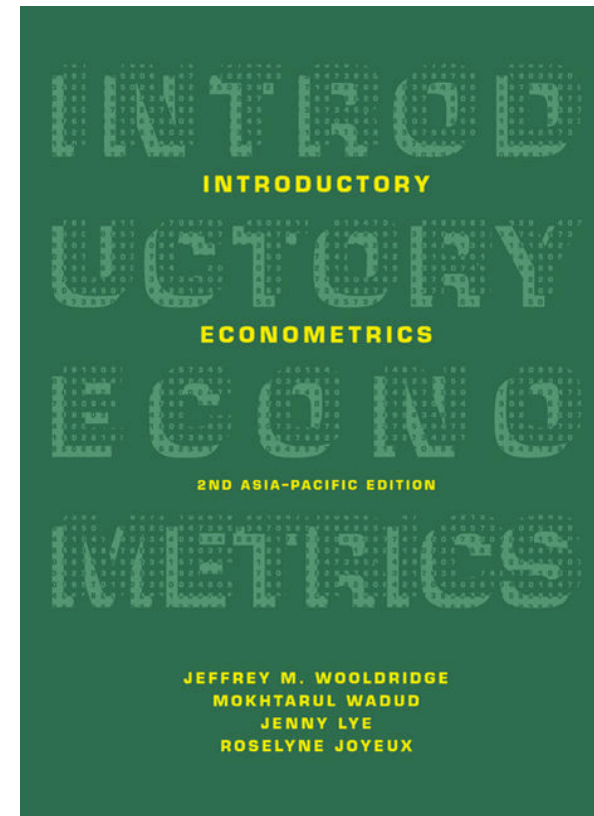
Python

- We'll cover using Python in next week's lecture (in Colab)
- Note: you'll need a google account

Textbook

Wooldridge et al. (2021)

Introductory Econometrics
(2nd Asia-Pacific ed.)



What is our goal?

a.k.a. demystifying learning outcomes

Learning outcomes

Learning outcomes

Specify, estimate & interpret a regression model

2

Learning outcomes

2

Specify, estimate & interpret a regression model

Answer: Did it work?



	Yields, enrolled vs. control	Total maize output
	(1)	(2)
Enrolled	295.43***	264.90***
	(37.23)	(85.73)
Control group mean	1128.39	1082.38
Observations	682	682

Learning outcomes

3

Summarise and interpret the estimation results, and draw valid inferences using hypothesis tests

What does this mean in practice?

Make sense of your results! Basically, you should know...

- what the numbers mean
- how strong the evidence is
- whether your conclusions are “real” or could be due to chance

Move from output → insight

Learning outcomes

4

Critically evaluate standard regression model assumptions & the consequences of violations

What does this mean in practice?

You should know and understand...

- what can go wrong when the assumptions underlying a regression model don't hold up
- what such violations mean for your results

Become informed consumers

Learning outcomes

5

Use econometric software to solve econometric problems

What does this mean in practice?

Use software to run econometric analyses



Build practical skills!



Each week, you should...

- 1 Read the assigned textbook chapter + any other assigned readings

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2 Watch the Essential Concepts videos (the lecture builds on them)

Watch: Essential concepts

What is econometrics?



What data do we use?



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- 2 Watch the Essential Concepts videos (the lecture builds on them)

3 Attend the live lecture

Review: this week's slides

What is econometrics?



What data do we use?



Intro to unit



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- 3 Attend the live lecture
- 4 Attempt the tutorial questions

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5 Attend the tutorial!

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- 4 Attempt the tutorial questions
- 5 Attend the tutorial!

6 Begin working on assessments

Each week, you should...

- 1 Read the assigned textbook chapter + any other assigned readings
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 - 3 Attend the live lecture
 - 4 Attempt the tutorial questions
 - 5 Attend the tutorial!
 - 6 Begin working on assessments
- 7** Ask for help if you need it!

Assessment

Quiz: 20%

Checks your grasp of core concepts and skills in applied econometrics, building on work from tutorials

Format: in-person quiz

Problem set: 30%

You'll draw on content from lectures & tutorials to apply econometrics to real problems

Format: short answer written submission

We're building something a little different: think choose-your-own-adventure, with data.

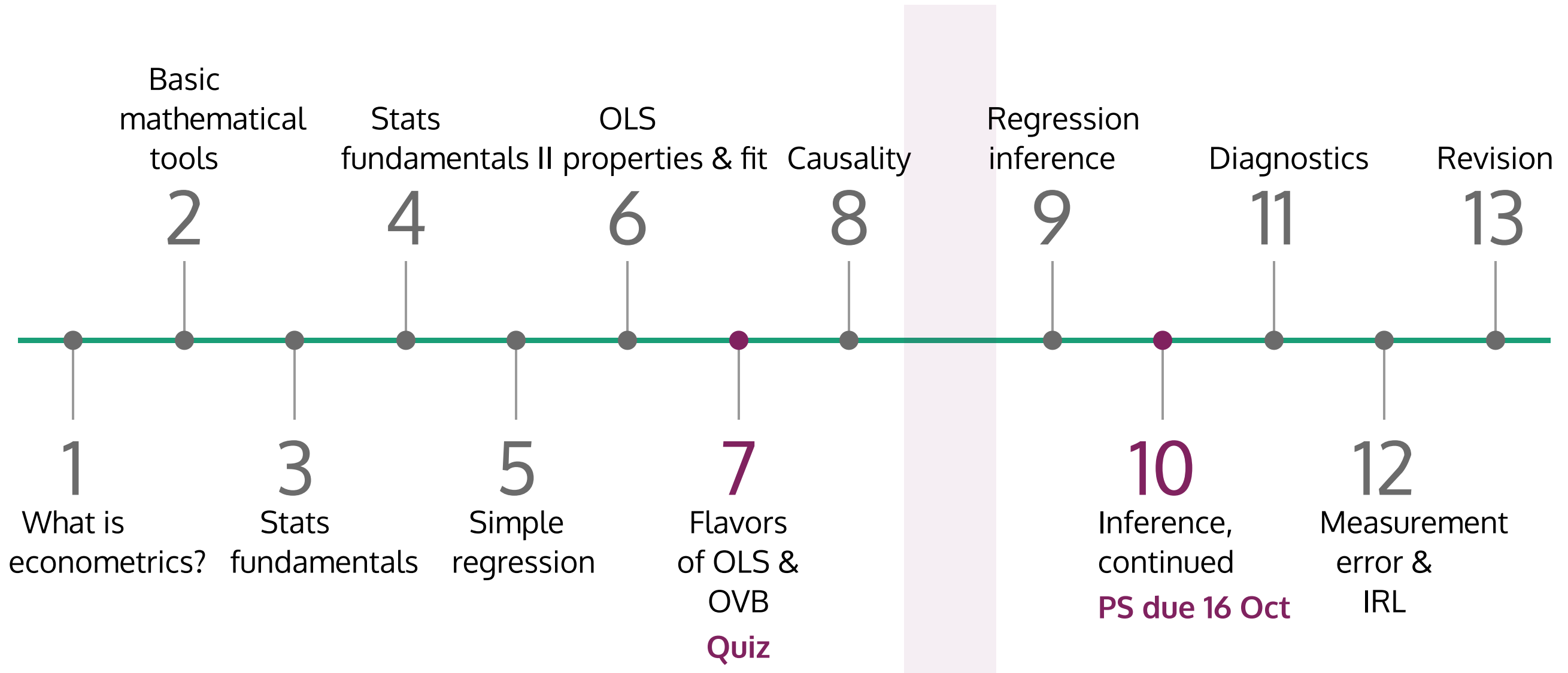
Watch this space!

Final exam: 50%

Format: 2-hour cumulative invigilated on-campus exam, held during Examination Period

The plan

Plan for the semester



Week 1

Essential concepts

- What is econometrics?
- What data do we use in econometrics?

Live lecture

- Intro to unit
- Salary prediction game

Tutorial

No tutorial in week 1

Week 2

Essential concepts

- Summation, proportions, percentages
- Partial derivatives, non-linear functions

Live lecture

- Intro to Python
- Write some code, bring laptops

Tutorial

- Get started with Python
- Understanding data types